

Solution Code - Module 3-Core Java - PalindromeChecker

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q16_PalindromeChecker.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q16_PalindromeChecker {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.print("Enter a string: ");
7 |
8 |         if (scanner.hasNextLine()) {
9 |             String input = scanner.nextLine();
10 |
11 |             // Clean the string: remove non-alphanumeric characters and convert to lowercase
12 |             String cleaned = input.replaceAll("[^a-zA-Z0-9]", "").toLowerCase();
13 |
14 |             // Reverse the cleaned string
15 |             String reversed = new StringBuilder(cleaned).reverse().toString();
16 |
17 |             System.out.println("Cleaned string: \"" + cleaned + "\"");
18 |             if (cleaned.equals(reversed)) {
19 |                 System.out.println("Result: The string is a palindrome.");
20 |             } else {
21 |                 System.out.println("Result: The string is not a palindrome.");
22 |             }
23 |         } else {
24 |             System.out.println("Error: No input found.");
25 |         }
26 |
27 |         scanner.close();
28 |     }
29 | }
```